

FMUSER FUQ-4454E QAM Modulator 8 Tuners-6 ASI-256 IP Inputs 4 RF Outputs DVB-C-EN300 429 ITU-T J.83 Mux Scrambling CATV Broadcasting



Product overview

The FUQ-4454E is a high-performance mux-scrambling QAM modulator featuring multiple input options including tuners, ASI, and IP ports, and 4 RF outputs. It processes signals through multiplexing, scrambling, and QAM modulation. With pluggable module design, UDP output support, and Web NMS, it ensures flexible control. Designed for modern CATV systems and broadcast applications, it delivers efficient signal distribution.

Key feature

Supports flexible input configurations with optional versions: Version 1 (8 tuners + 6 ASI + 256 IP) or Version 2 (4 tuners with CAM + 6 ASI)

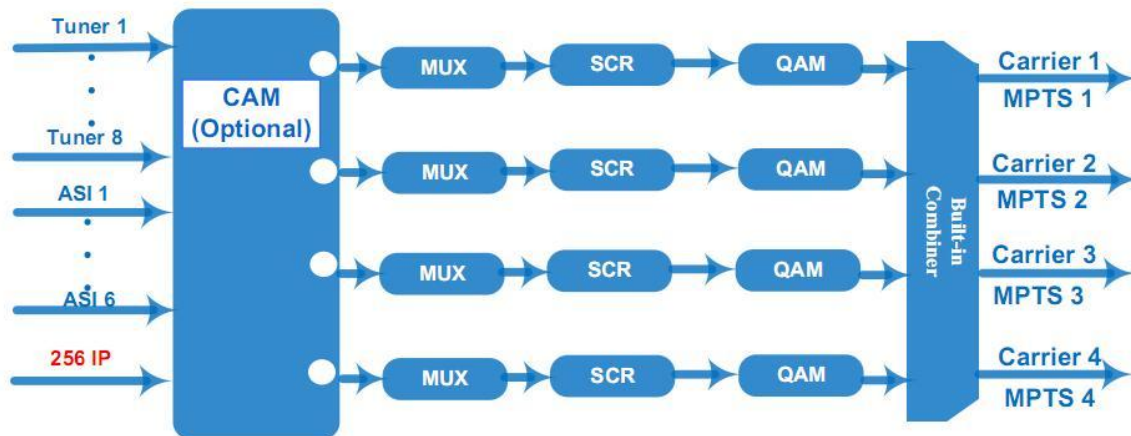
Integrates 4-channel multiplexing, scrambling, and QAM modulation compliant with DVB-C and ITU-T J.83A/B/C

Delivers superior RF output with MER ≥ 40 dB and adjustable frequency/level

Provides advanced output options including 4MPTS over UDP and BISS descrambling (Version 2)

Features user-friendly management via LCD display, keyboard, and Web NMS

Working Principle



Key feature

- * Multiple Input ports optional:
 - 8 DVB-C/T/(T)/ISDBT/S/S2/S2X/ATSC FTA Tuner + 6 ASI input ports+256 IP input (Version 1)
 - 4 DVB-C/S/S2/ATSC Tuner with CAM+6 ASI input ports (Version 2)
- * Support 4 channels multiplexing+4 channels scrambling+4 channels QAM modulating
- * Fully support DVB-C (EN300 429) and ITU-T J.83A/B/C
- * Fully support DVB general scrambling system description ETR289, simulcrypt standard ETSI 101 197 and ETSI 103 197
- * Support accurate PCR adjusting
- * Support PSI/SI editing and inserting
- * Excellent RF output performance index, MER≥40db
- * Support 4*MPTS output over UDP
- * 4 CAM decrypt multiple programs from 4 Tuner input (Optional)
- * Support BISS descrambling(only for Version 2)
- * Support LCD display and keyboard
- * Support Web management, Updates via web and USB

Specifications

Input	8 DVB-C/ T/(T)/ISD BT/S/S2/ S2X/ATS C FTA Tuner + 6 ASI input+256 IP input -- ----Versio n 1	8 DVB-C/ T/(T)/ISD BT/S/S2/ S2X/ATS C FTA Tuner + 6 ASI input+256 IP input -- ----Versio n 1	8 DVB-C/ T/(T)/ISD BT/S/S2/ S2X/ATS C FTA Tuner + 6 ASI input+256 IP input -- ----Versio n 1	8 DVB-C/ T/(T)/ISD BT/S/S2/ S2X/ATS C FTA Tuner + 6 ASI input+256 IP input -- ----Versio n 1	8 DVB-C/ T/(T)/ISD BT/S/S2/ S2X/ATS C FTA Tuner + 6 ASI input+256 IP input -- ----Versio n 1	8 DVB-C/ T/(T)/ISD BT/S/S2/ S2X/ATS C FTA Tuner + 6 ASI input+256 IP input -- ----Versio n 1	8 DVB-C/ T/(T)/ISD BT/S/S2/ S2X/ATS C FTA Tuner + 6 ASI input+256 IP input -- ----Versio n 1
-------	--	--	--	--	--	--	--

Input	4 DVB-C/ S/S2/ATS C Tuner with CAM + 6 ASI ----- Version 2	4 DVB-C/ S/S2/ATS C Tuner with CAM + 6 ASI ----- Version 2	4 DVB-C/ S/S2/ATS C Tuner with CAM + 6 ASI ----- Version 2	4 DVB-C/ S/S2/ATS C Tuner with CAM + 6 ASI ----- Version 2	4 DVB-C/ S/S2/ATS C Tuner with CAM + 6 ASI ----- Version 2	4 DVB-C/ S/S2/ATS C Tuner with CAM + 6 ASI ----- Version 2	4 DVB-C/ S/S2/ATS C Tuner with CAM + 6 ASI ----- Version 2
Tuner Section	For Version 1 ■ Multi- mode tuners sw itchable (New Type)	DVB-C	Input Fre quency	Input Fre quency	60 ~890 MHz	60 ~890 MHz	60 ~890 MHz
Tuner Section	For Version 1 ■ Multi- mode tuners sw itchable (New Type)	DVB-C	Standard	Standard	J.83A, J.83C	J.83A, J.83C	J.83B
Tuner Section	For Version 1 ■ Multi- mode tuners sw itchable (New Type)	DVB-C	Constella tion	Constella tion	16/32/64/ 128/256 QAM	16/32/64/ 128/256 QAM	64/256 QAM
Tuner Section	For Version 1 ■ Multi- mode tuners sw itchable (New Type)	DVB-T/(T)	Input Fre quency	Input Fre quency	60 ~890 MHz	60 ~890 MHz	60 ~890 MHz

Tuner Section	For Version 1 ■ Multi-mode tuners sw itchable (New Type)	DVB-T/(T)	Bandwidth	Bandwidth	5/6/7/8 M	5/6/7/8 M	5/6/7/8 M
Tuner Section	For Version 1 ■ Multi-mode tuners sw itchable (New Type)	ISDB-T	Input Frequency	Input Frequency	60 ~890 MHz	60 ~890 MHz	60 ~890 MHz
Tuner Section	For Version 1 (New Type)	DVB-S	Input Frequency	Input Frequency	950~2150MHz	950~2150MHz	950~2150MHz
Tuner Section	For Version 1 (New Type)	DVB-S	Symbol rate	Symbol rate	0.5~45MSPS	0.5~45MSPS	0.5~45MSPS
Tuner Section	For Version 1 (New Type)	DVB-S	Signal Strength	Signal Strength	- 65~-25dBm	- 65~-25dBm	- 65~-25dBm
Tuner Section	For Version 1 (New Type)	DVB-S	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Tuner Section	For Version 1 (New Type)	DVB-S	Constellation	Constellation	QPSK	QPSK	QPSK
Tuner Section	For Version 1 (New Type)	DVB-S	Max input bitrate	Max input bitrate	≤120 Mbps	≤120 Mbps	≤120 Mbps

Tuner Section	For Version 1 (New Type)	DVB-S2X	Input Frequency	Input Frequency	950~2150MHz	950~2150MHz	950~2150MHz
Tuner Section	For Version 1 (New Type)	DVB-S2X	Symbol rate	Symbol rate	QPSK/8PSK/16APSK: 0.5~45 Msps 32APSK: 0.5~40Mbps;	QPSK/8PSK/16APSK: 0.5~45 Msps 32APSK: 0.5~40Mbps;	QPSK/8PSK/16APSK: 0.5~45 Msps 32APSK: 0.5~40Mbps;
Tuner Section	For Version 1 (New Type)	DVB-S2X	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10
Tuner Section	For Version 1 (New Type)	DVB-S2X	Constellation	Constellation	QPSK, 8PSK, 16APSK, 32APSK	QPSK, 8PSK, 16APSK, 32APSK	QPSK, 8PSK, 16APSK, 32APSK
Tuner Section	For Version 1 (New Type)	DVB-S2X	Max input bitrate	Max input bitrate	≤120 Mbps	≤120 Mbps	≤120 Mbps

Tuner Section	For Version 1 (New Type)	DVB-S2X	Input Frequency	Input Frequency	950~2150MHz	950~2150MHz	950~2150MHz
Tuner Section	For Version 1 (New Type)	DVB-S2X	Symbol rate	Symbol rate	QPSK/8PSK/16APSK: 0.5~45 Msps 8APSK: 0.5~40Msps 32APSK: 0.5~40Msps	QPSK/8PSK/16APSK: 0.5~45 Msps 8APSK: 0.5~40Msps 32APSK: 0.5~40Msps	QPSK/8PSK/16APSK: 0.5~45 Msps 8APSK: 0.5~40Msps 32APSK: 0.5~40Msps

Tuner Section	For Version 1 (New Type)	DVB-S2X	FEC	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36 , 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36 , 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36 , 2/3-L, 25/36, 13/18, 7/9, 77/90 32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9
Tuner Section	For Version 1 (New Type)	DVB-S2X	Constellation	Constellation	QPSK, 8PSK, 8APSK, 16APSK, 32APSK	QPSK, 8PSK, 8APSK, 16APSK, 32APSK	QPSK, 8PSK, 8APSK, 16APSK, 32APSK
Tuner Section	For Version 1 (New Type)	DVB-S2X	Max input bitrate	Max input bitrate	≤120 Mbps	≤120 Mbps	≤120 Mbps

Tuner Section	For Version 2	DVB-C	Standard	Standard	J.83A(DV B-C), J.83B, J.83C	J.83A(DV B-C), J.83B, J.83C	J.83A(DV B-C), J.83B, J.83C
Tuner Section	For Version 2	DVB-C	Input Frequency	Input Frequency	30~1000 Mhz	30~1000 Mhz	30~1000 Mhz
Tuner Section	For Version 2	DVB-C	Constellation	Constellation	16/32/64/128/256 QAM	16/32/64/128/256 QAM	16/32/64/128/256 QAM
Tuner Section	For Version 2	DVB-S	Input Frequency	Input Frequency	950~2150MHz	950~2150MHz	950~2150MHz
Tuner Section	For Version 2	DVB-S	Symbol rate	Symbol rate	2-45Msps	2-45Msps	2-45Msps
Tuner Section	For Version 2	DVB-S	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Tuner Section	For Version 2	DVB-S	Signal Strength	Signal Strength	-65~-25dBm	-65~-25dBm	-65~-25dBm
Tuner Section	For Version 2	DVB-S	Constellation	Constellation	QPSK	QPSK	QPSK
Tuner Section	For Version 2	DVB-S2	Input Frequency	Input Frequency	950~2150MHz	950~2150MHz	950~2150MHz
Tuner Section	For Version 2	DVB-S2	Symbol rate	Symbol rate	QPSK/8PSK 1~45 Msps	QPSK/8PSK 1~45 Msps	QPSK/8PSK 1~45 Msps
Tuner Section	For Version 2	DVB-S2	FEC	FEC	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10
Tuner Section	For Version 2	DVB-S2	Constellation	Constellation	QPSK, 8PSK	QPSK, 8PSK	QPSK, 8PSK
Tuner Section		ATSC	Input Frequency	Input Frequency	54~858MHz	54~858MHz	54~858MHz
Tuner Section		ATSC	Bandwidth	Bandwidth	6M	6M	6M
Tuner Section		ATSC	Constellation	Constellation	8VSB	8VSB	8VSB

Multiplexing	Maximum PID Remapping	128per input channel	128per input channel	128per input channel	128per input channel	128per input channel	128per input channel
Multiplexing	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically
Scrambling	Maximum simulcrypt CA	4	4	4	4	4	4
Scrambling	Standard	ETSI 101 197, ETSI 103 197	ETSI 101 197, ETSI 103 197	ETSI 101 197, ETSI 103 197	ETSI 101 197, ETSI 103 197	ETSI 101 197, ETSI 103 197	ETSI 101 197, ETSI 103 197
Scrambling	Connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection
Modulation	QAM Channel	4	4	4	4	4	4
Modulation	Standard	EN300 429/ITU-T J.83A/B/C	EN300 429/ITU-T J.83A/B/C	EN300 429/ITU-T J.83A/B/C	EN300 429/ITU-T J.83A/B/C	EN300 429/ITU-T J.83A/B/C	EN300 429/ITU-T J.83A/B/C
Modulation	MER	≥40db	≥40db	≥40db	≥40db	≥40db	≥40db
Modulation	RF frequency	30~999M Hz, 1KHz step	30~999M Hz, 1KHz step	30~999M Hz, 1KHz step	30~999M Hz, 1KHz step	30~999M Hz, 1KHz step	30~999M Hz, 1KHz step

Modulation	RF output level	-10~0dbm, 0.1db step ----- Version 1 -20~0dbm, 0.1db step ----- Version 2	-10~0dbm, 0.1db step ----- Version 1 -20~0dbm, 0.1db step ----- Version 2	-10~0dbm, 0.1db step ----- Version 1 -20~0dbm, 0.1db step ----- Version 2	-10~0dbm, 0.1db step ----- Version 1 -20~0dbm, 0.1db step ----- Version 2	-10~0dbm, 0.1db step ----- Version 1 -20~0dbm, 0.1db step ----- Version 2	-10~0dbm, 0.1db step ----- Version 1 -20~0dbm, 0.1db step ----- Version 2
Modulation	Symbol Rate	5.0Msps~9.0Msps, 1ksps stepping	5.0Msps~9.0Msps, 1ksps stepping	5.0Msps~9.0Msps, 1ksps stepping	5.0Msps~9.0Msps, 1ksps stepping	5.0Msps~9.0Msps, 1ksps stepping	5.0Msps~9.0Msps, 1ksps stepping
Modulation		J.83A	J.83A	J.83B	J.83B	J.83C	J.83C
Modulation	Constellation	16/32/64/128/256QAM	16/32/64/128/256QAM	64/256QAM	64/256QAM	64/256QAM	64/256QAM
Modulation	Bandwidth	8M	8M	6M	6M	6M	6M
System	Local interface	LCD + control buttons	LCD + control buttons	LCD + control buttons	LCD + control buttons	LCD + control buttons	LCD + control buttons
System	Remote management	Web NMS	Web NMS	Web NMS	Web NMS	Web NMS	Web NMS
System	Stream Out	4*MPTS over UDP out (RJ45, 1000M, Unicast/Multicast)	4*MPTS over UDP out (RJ45, 1000M, Unicast/Multicast)	4*MPTS over UDP out (RJ45, 1000M, Unicast/Multicast)	4*MPTS over UDP out (RJ45, 1000M, Unicast/Multicast)	4*MPTS over UDP out (RJ45, 1000M, Unicast/Multicast)	4*MPTS over UDP out (RJ45, 1000M, Unicast/Multicast)
System	Language	English and Chinese	English and Chinese	English and Chinese	English and Chinese	English and Chinese	English and Chinese
System	Software Upgrading	Web and USB	Web and USB	Web and USB	Web and USB	Web and USB	Web and USB
General	Dimension (W*D*H)	482mm×300mm×44.5mm	482mm×300mm×44.5mm	482mm×300mm×44.5mm	482mm×300mm×44.5mm	482mm×300mm×44.5mm	482mm×300mm×44.5mm

General	Temperat ure	0~45■(O peration) ; -20~80 ■(Storag e)	0~45■(O peration) ; -20~80 ■(Storag e)	0~45■(O peration) ; -20~80 ■(Storag e)	0~45■(O peration) ; -20~80 ■(Storag e)	0~45■(O peration) ; -20~80 ■(Storag e)	0~45■(O peration) ; -20~80 ■(Storag e)
General	Power	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ	AC 100V ±1050/60 Hz; AC 2 20V±10% , 50/60HZ

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: [+8613922702227](https://wa.me/8613922702227)